

Container-based high-performance computing for modeling large-scale biological neural networks

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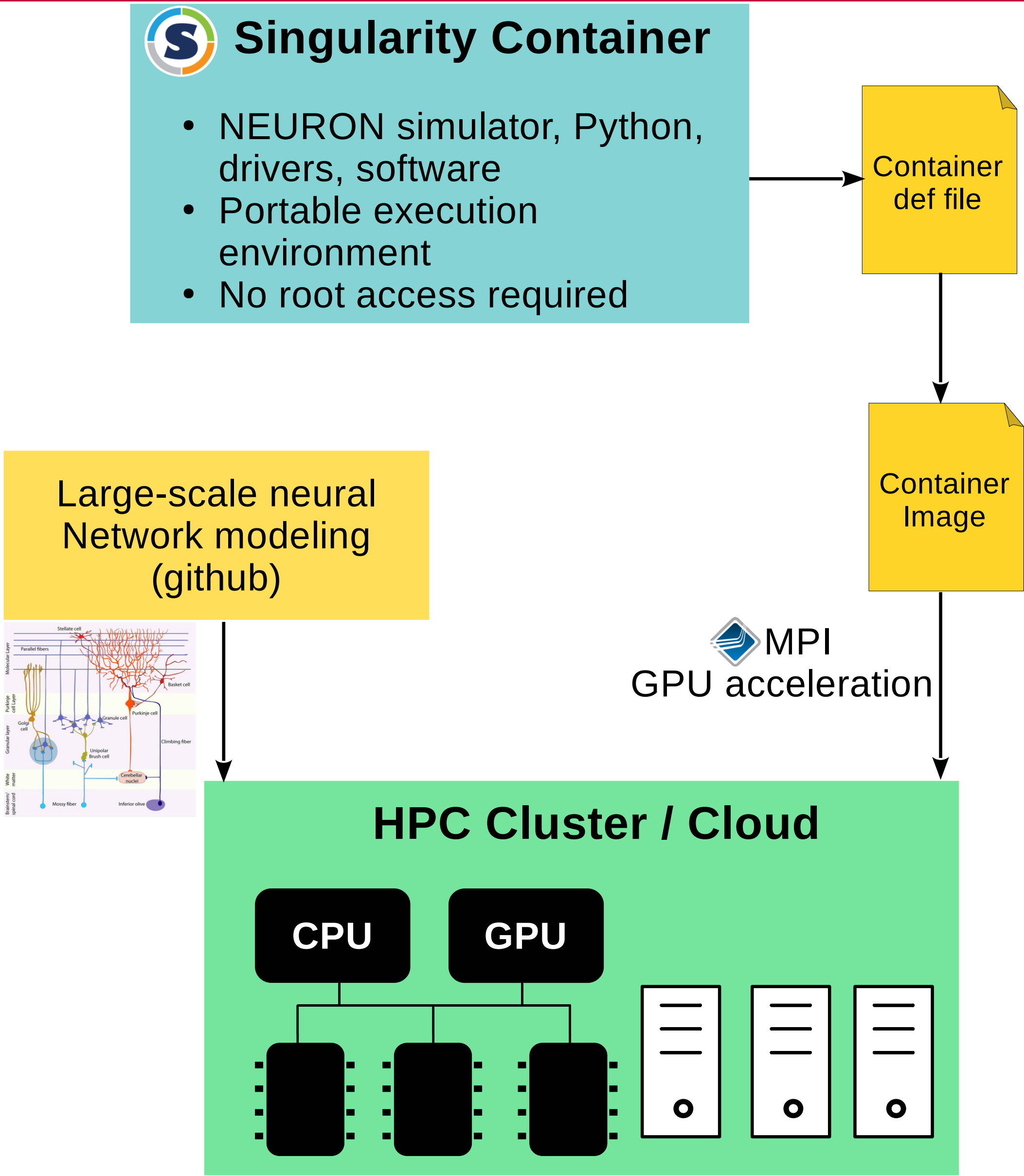
Benchmarking Data and Network*: Cerebellar Mossy Fiber → Granule Cell

Introduction

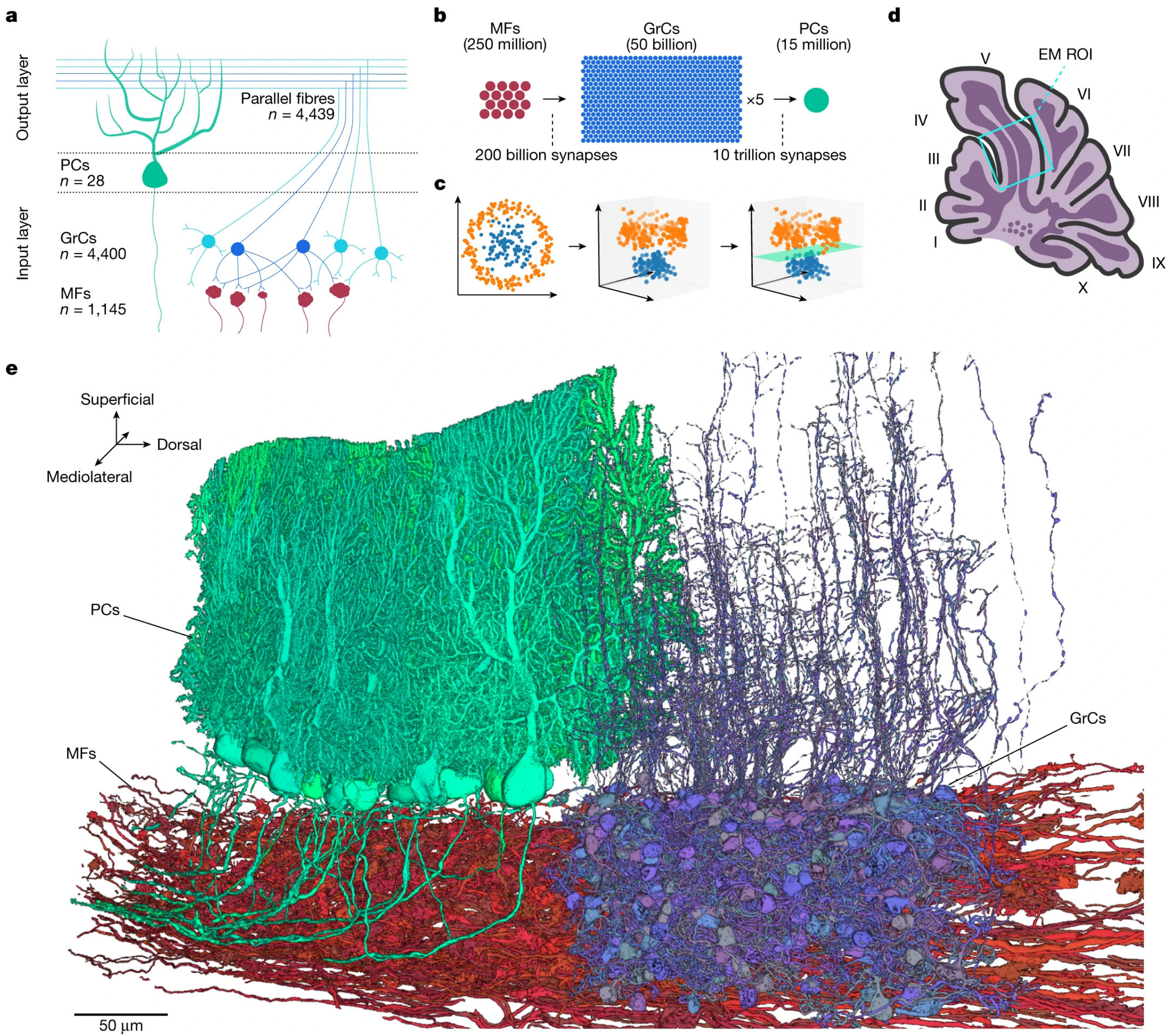
1. Modeling large-scale, biophysically detailed neural networks increasingly relies on high-performance computing (HPC).
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to software incompatibilities and complex dependencies
3. We address this issue using singularity containers that encapsulate the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on shared HPC systems.
4. We give a potential practical demonstrations using our cerebellar mossy fiber - granule cell network simulations.

Containerization for Scientific Computing

Our Computational Framework



A slight change in mask contrast induces blindsight-like behavioral performance.



* Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. Nature 613, 543–549 (2023)

Number of MF: 85600

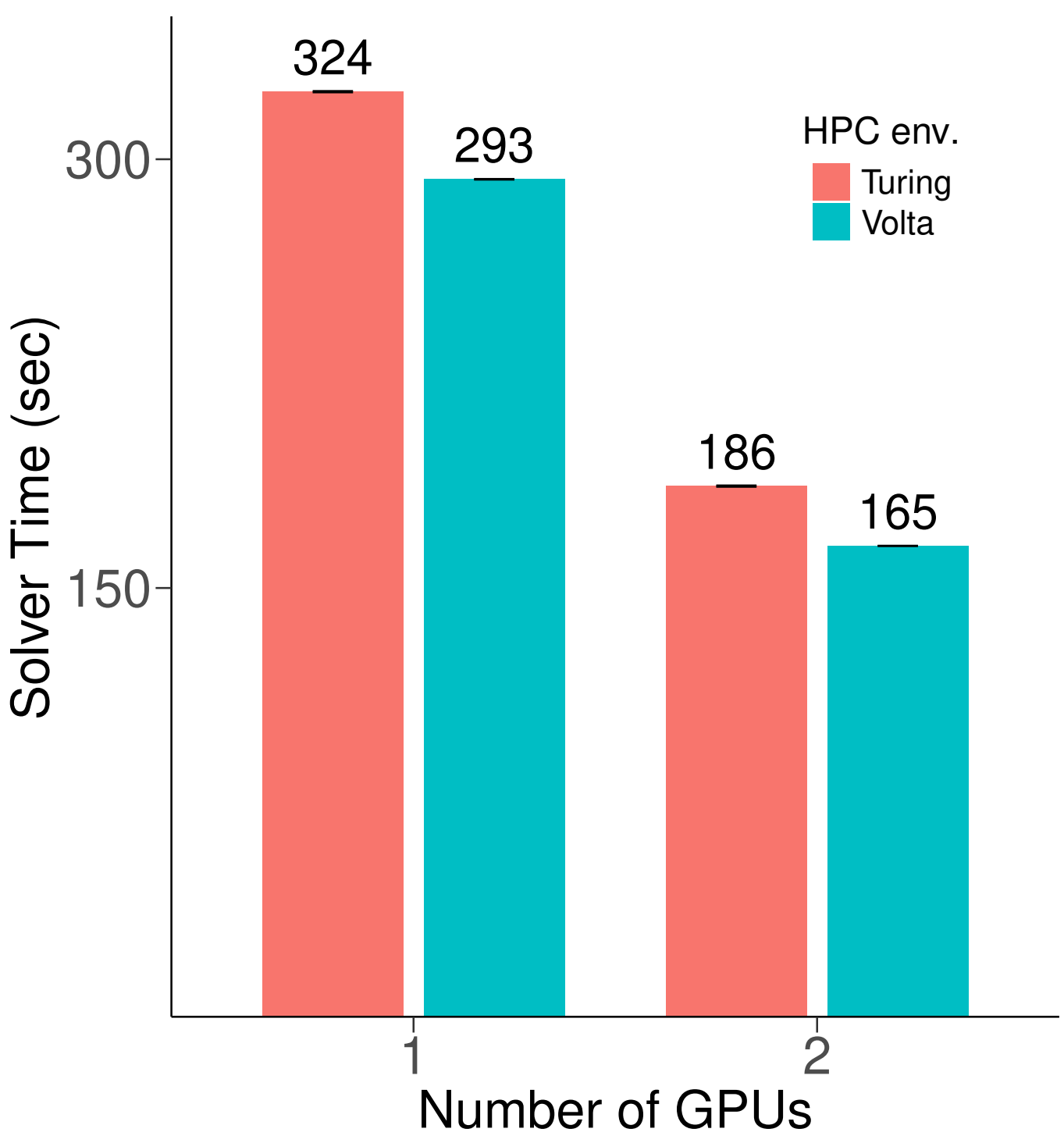
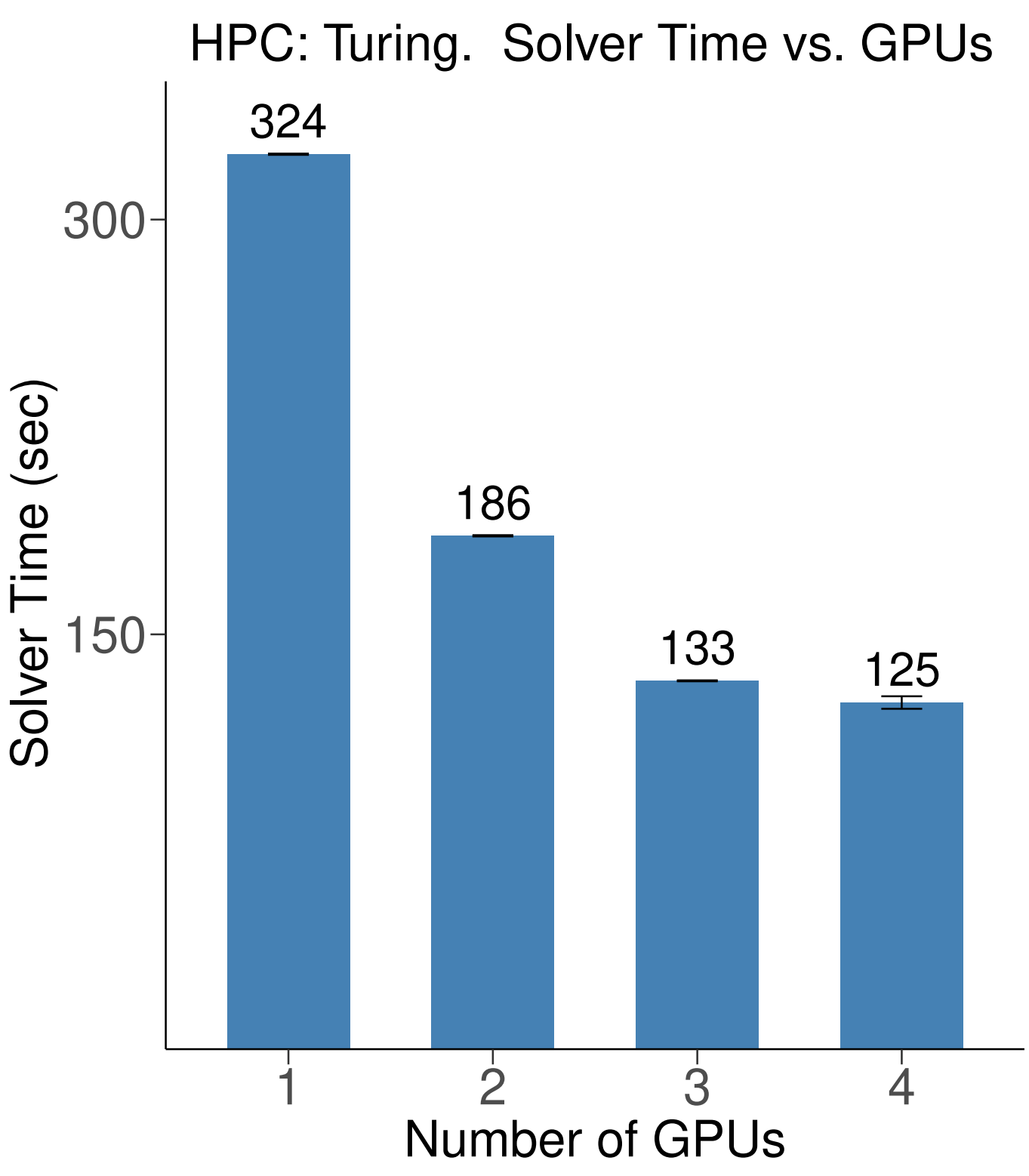
Number of GRC: 314000

Number of compartments: 942000

Number of synapses: 990960

Simulation time: 2 sec

Performance Across HPC Environment



Acknowledgment

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